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Chapter: Database Keys

Natural Key

A Natural Key is a key that is derived from real-world business data and naturally exists outside the database.

It already has meaning in the business domain and can uniquely identify an entity.

Pros

- already exists
- meaningful
- no additional column needed

...are its pros

Cons

- value or business logic may change
- long values
- privacy
- external dependancy

...are cons

Natural Keys may change over time, can be long, may expose sensitive business or personal information, and are tied to business rules that can evolve.

Surrogate Key

Definition

A Surrogate Key is an artificially generated key with no business meaning, created solely to uniquely identify a row.

A Surrogate Key:

- Is generated by the system.

- Has no business meaning.
- Is usually immutable (does not change).
- Is unique.
- Is typically short.
- Is commonly used as the Primary Key.

Pros

stable , small , faster joins , secure

Cons

however no business meaning and additional columns is required while business logic still need enforcement

Generating Surrogate Keys

Surrogate Keys can be generated in different ways.

- **Auto Increment Integer**
  1
  2
  3
  4
- **UUID**
  550e8400-e29b-41d4-a716-446655440000
- **snowflake ids**
  unique 64-bit numeric IDs that encode information like time and machine identifiers.

UUID (Universally Unique Identifier)

Instead of integers, some systems generate UUIDs.

Example
`550e8400-e29b-41d4-a716-446655440000`

Every generated UUID is designed to be globally unique with an extremely low probability of collision.

UUID? Useful when

- Multiple servers generate IDs independently.
- Offline applications need to create records.
- Distributed databases are used.

When to Use Integer IDs vs UUIDs

When to Use Integer IDs	When to Use UUIDs
• Single database server • Most web applications • Banking systems • College projects • Inventory systems	• Distributed databases • Microservices • Offline synchronization • Multi-region deployments

"Surrogate Keys replace Natural Keys."

❌ False.

Natural Keys often remain in the table with a UNIQUE constraint to enforce business rules.